

Probability

Conditional probability problem

1) A die is tossed, if the number is odd on the face, what is the probability that it is prime.

Solⁿ:- Sample space, $S = \{1, 2, 3, 4, 5, 6\}$

Let E be the event that denotes getting an odd number when a die is tossed. Then

$$E = \{1, 3, 5\} \Rightarrow P(E) = \frac{3}{6} = \frac{1}{2}$$

F be the event that denotes getting a prime number when a die is tossed. Then

$$F = \{2, 3, 5\} \Rightarrow P(F) = \frac{3}{6} = \frac{1}{2}$$

$$E \cap F = \{3, 5\} \Rightarrow P(E \cap F) = \frac{2}{6} = \frac{1}{3}$$

$$P(F|E) = \frac{P(E \cap F)}{P(E)} = \frac{\frac{1}{3}}{\frac{1}{2}} = \frac{2}{3}$$

2. If A and B are two independent events of S such that $P(A) = \frac{1}{3}$, $P(B) = \frac{1}{4}$, $P(A \cup B) = \frac{1}{2}$, then find
i) $P(A|B)$ ii) $P(B|A)$

Solⁿ:- Since A and B are independent events,
 $P(A \cap B) = P(A) \cdot P(B)$
 $= \frac{1}{3} \times \frac{1}{4} = \frac{1}{12}$

$$\text{i) } P(A|B) = \frac{P(A \cap B)}{P(B)} = \frac{\frac{1}{12}}{\frac{1}{4}} = \frac{4}{12} = \frac{1}{3}$$

$$\text{ii) } P(B|A) = \frac{P(A \cap B)}{P(A)} = \frac{\frac{1}{12}}{\frac{1}{3}} = \frac{3}{12} = \frac{1}{4}$$

Total probability theorem, Baye's theorem

1) A person has undertaken a construction job. The probabilities are 0.65 that there will be a strike, 0.80 that the construction job will be completed on time if there is no strike, and 0.32 that the construction job will be completed on time if there is a strike. Determine the probability that the construction job will be completed on time.

Solⁿ:- let A be the event that the construction job will be completed on time. B be the event that there will be a strike.

We have to find $P(A)$

$$P(B) = P(\text{strike}) = 0.65$$

$$\Rightarrow P(B') = 1 - P(B) = 1 - 0.65 = 0.35$$

$$P(A|B) = 0.32 \quad P(A|B') = 0.80$$

Here B and B' form a partition of the sample space S.

Therefore by theorem on total probability

$$\begin{aligned} P(A) &= P(B) P(A|B) + P(B') P(A|B') \\ &= 0.65 \times 0.32 + 0.35 \times 0.80 \\ &= 0.488 \end{aligned}$$

2) Suppose 3 companies x, y, z produce TVs. x produces twice as many as y , while y and z produce same number. It is known that 2% of x , 2% of y , 4% of z are defected. All the TVs are produced are put into 1 shop and then 1 TV is chosen at random. What is the probability that the TV is defected. Suppose a TV chosen is defective, what is the probability that this TV is produced by company x .

Solⁿ:- let X denote the event
tv is produced by company x
let Y denote the event
tv is produced by company y
let Z denote the event
tv is produced by company z .
let D denote the event
tv is defected.

Given $P(X) = 0.5$, $P(Y) = 0.25$, $P(Z) = 0.25$

$$P(D|X) = 0.02$$

$$P(D|Y) = 0.02$$

$$P(D|Z) = 0.04$$

$$\begin{aligned} \text{i) } P(D) &= P(X) P(D|X) + P(Y) P(D|Y) \\ &\quad + P(Z) P(D|Z) \end{aligned}$$

$$\begin{aligned} &= 0.5(0.02) + 0.25(0.02) + 0.25(0.04) \\ &= 0.025 \end{aligned}$$

The probability that the tv is detected is 0.025

$$\begin{aligned} \text{ii) } P(X|D) &= \frac{P(X) P(D|X)}{P(X) P(D|X) + P(Y) P(D|Y) \\ &\quad + P(Z) P(D|Z)} \end{aligned}$$

$$\begin{aligned} &= \frac{0.5(0.02)}{0.5(0.02) + 0.25(0.02) + 0.25(0.04)} \\ &= 0.4 \end{aligned}$$

3) There are 3 boxes, the first one containing 1 white, 2 red and 3 black balls, the 2nd one containing 2W, 3R & 1 B ball and the 3rd one containing 3W, 1R and 2B balls. A box is chosen at random. and from it 2 balls are drawn at random. One ball is red and the other white. What is the probability that they come from the second box?

Solⁿ:- Let B_1 be the event of choosing ball from box 1

B_2 be the event of choosing ball from box 2

B_3 be the event of choosing ball from box 3.

$$P(B_1) = \frac{1}{3}, \quad P(B_2) = \frac{1}{3}, \quad P(B_3) = \frac{1}{3}$$

Let A be the event that one red ball and one white ball is drawn.

We have to find $P(B_2|A)$

Total no of ways to choose two balls from six balls in any box is

$$\binom{6}{2} = 15$$

For box 1 (1W, 2R, 3B):-

The no of ways to choose one white and one red ball is

$$\binom{1}{1} \times \binom{2}{1} = 2$$

$$\therefore P(A|B_1) = \frac{2}{15}$$

For box 2 (2W, 3R, 1B):-

The no of ways to choose one white and one red ball is

$$\binom{2}{1} \times \binom{3}{1} = 6$$

$$\therefore P(A|B_2) = \frac{6}{15} = \frac{2}{5}$$

For box 3 (2W, 3R, 1B):-

The no of ways to choose one white and one red ball is

$$\binom{2}{1} \times \binom{3}{1} = 6$$

$$\therefore P(A|B_3) = \frac{6}{15} = \frac{2}{5}$$

$$P(A) = P(A|B_1) P(B_1) + P(A|B_2) P(B_2) + P(A|B_3) P(B_3)$$

$$P(A) = \frac{2}{15} \times \frac{1}{3} + \frac{2}{5} \times \frac{1}{3} + \frac{2}{5} \times \frac{1}{3}$$

$$= \frac{11}{45}$$

By Baye's theorem

$$P(B_2|A) = \frac{P(A|B_2) \cdot P(B_2)}{P(A)}$$

$$= \frac{\left(\frac{6}{15}\right) \left(\frac{1}{3}\right)}{\left(\frac{11}{45}\right)} = \frac{6}{11}$$

4) One percent of the population suffers from a certain disease. There is blood test for this disease, and it is 99% accurate, in other words, the probability that gives the correct answer is 0.99, regardless of whether the person is sick or healthy. A person takes the blood test, and the result says that he has the disease. The probability that he actually has the disease, is?

A be the event of having the disease
B be the event of testing +ve.

$$P(A) = 0.01$$

$$\begin{aligned} P(B) &= P(B|A)P(A) + P(B|A')P(A') \\ &= 0.01 \times 0.99 + 0.01 \times 0.99 \\ &= 0.0198 \end{aligned}$$

$$P(A \cap B) = 0.99 \times 0.01$$

$$P(A|B) = \frac{P(A \cap B)}{P(B)} = \frac{0.99 \times 0.01}{0.0198} = \underline{\underline{0.5}}$$

Random variable

Ex:- Consider the random experiment of tossing a die.

$$S = \{1, 2, 3, 4, 5, 6\}$$

Let $X = S \rightarrow R$ be a function defined by $X(S) = 10S$

$$X(1) = 10, \quad X(2) = 20, \quad \dots, \quad X(6) = 60$$

X is a Random variable.

Suppose $Y = S \rightarrow R$ is a function defined by

$$X(S) = 0 \quad \text{if } S \text{ is odd}$$

$$X(S) = 1 \quad \text{if } S \text{ is even.}$$

Then X takes value either 0 or 1.

$\therefore X$ is a Random variable.